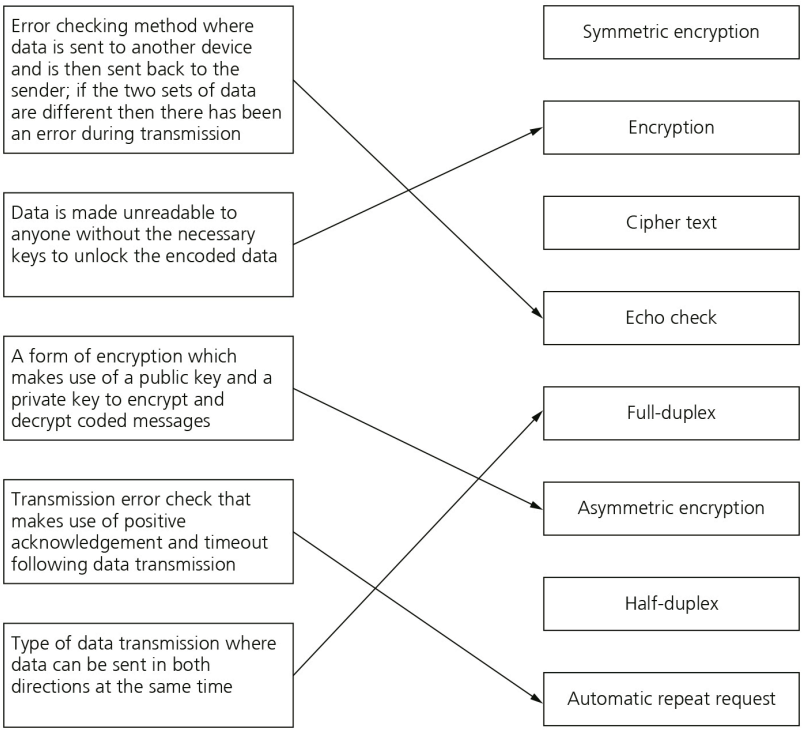
10



11

- asymmetric encryption
- public
- private
- document
- algorithm
- matching pairs
- mathematically
- public key
- public key
- encrypted
- matching private
- document
- decrypt

3 Hardware

1

- step 1: 4
- step 2: 1
- step 3: 2
- step 4: given in flowchart
- step 5: 5
- step 6: 3

Please note: step 5 (increment PC) could also be shown in steps 2 or 3 in the flowchart.

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2 a address bus, data bus and control bus

b i MAR: 1111 1110 MDR: 1010 0111

ii MAR: 1000 0100 MDR: 1100 1101

iii

Address	Contents
1000 0000	0111 1100
1000 0001	1000 0011
1000 0010	1111 1101
1000 0011	0111 1110
1000 0100	1100 1101
↓	↓
1111 1100	**0011 1100**
1111 1101	0110 0011
1111 1110	1010 0111
1111 1111	1111 0011

The greyed cell in the diagram in the Workbook should contain 00111100.

c CIR: current instruction register – stores the current instructions being decoded and executed

MAR: memory address register – stores the address of the memory location currently being read from/written to

MDR: memory data register – stores data which has just been read from memory/data and is about to be written to memory

PC: program counter – stores the address where the next instruction to be read can be found

3 a i Clock speeds are measured in GHz, which is the vibrational frequency of the clock which sends out pulses along the control bus to synchronise operations.

ii A temporary memory location using static RAM to hold data/instructions that are frequently used by the processor; a larger cache memory increases processor performance.

b Increasing clock speed can improve performance; however, it can lead to overclocking, which can cause operations to be unsynchronised, which can cause serious overheating of the CPU. Use of more cores can improve processor performance (for example, dual core and quad core). However, doubling the number of cores doesn't necessarily double the performance because the CPU needs to communicate with each core, reducing the overall performance.

4 a i a combination of hardware and software designed to carry out a specific set of instructions

ii In any order:

- small in size and therefore easy to fit into devices
- relatively cheap to manufacture
- dedicated to a single task which allows for a simple interface and OS
- consumes very little power

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iii Three from:

- set top boxes – to allow the recording and playback of TV programmes
- security systems – use sensors to check for intruders and send a warning signal if an intruder is detected
- lighting systems – automatic control of lighting using light sensors, for example, time of day, whether the room is occupied, light intensity changes, etc.
- vending machines – to automatically supply customers with an item from a machine; they use a keypad, sensors and a number of actuators to supply the item
- motor vehicles – they control/monitor the functions of many of the components in a car
- washing machines – automatically control the wash cycle

b i Gaming devices involve sensors to give a degree of realism to games:

- accelerometers (these measure acceleration and deceleration and therefore measure and respond to tilting the gaming device forward/backward and side to side)
- proximity sensors (used in smart touch pads; here electrodes are embedded in touch pads which can detect hand/finger position thus increasing user movements and user awareness)
- embedded accelerometers and proximity sensors (together with a microcontroller) in games consoles allow increased human interaction with the game – this allows players to take actions which simulate real events happening, giving a more immersive games experience.

ii
- automatic updates are sent via a cellular network
- automatic updates are received by plugging a device into a computer or via a WiFi/Bluetooth connection

5 a A: 0 1 1 1 1 0 1

B: 0 1 0 1 1 1 1

C: 0 1 1 0 0 0 1

b Inputs:

- scanner – to read the barcode
- keypad – to enter the barcode number if the barcode fails to scan
- card reader – to allow payments

Outputs:

- monitor – to show an item's description and/or price
- loudspeaker – to give 'beeping' sounds to indicate that the item has been read correctly or that there was an error in reading the barcode
- printer – to produce an itemised bill and customised vouchers

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c Manager, two from:

- easier to change the prices on stock
- more up-to-date sales information can be automatically generated
- no need to price each item individually
- automatic stock control
- customer's buying habits can be easily checked (by linking purchases to a loyalty card)

Customer, two from:

- faster checkouts
- fewer errors in the bills
- itemised billing
- cost savings passed on to customers
- less chance of food being out of date

6 a

- adjust the shutter speed
- focus the image automatically
- operate the flash automatically
- adjust the aperture size
- adjust the size of the image
- 'red eye'

b Note: don't accept answers such as 'taking a photograph of a scene'.

Two from:

- reading QR codes using a smart phone/tablet
- robots and other devices where information about the surroundings is needed (e.g. autonomous vehicles or reconnaissance drones)
- diagnosis of patients (e.g. dental work for easier detection of problems)
- security systems (e.g. use of infrared sensors to trigger a camera to take a photo or move the camera)
- video conferencing and other forms of conferencing
- producing virtual reality tours
- producing data for use in satellite navigation systems and online 'maps'
- augmented reality (e.g. when selling cosmetics, the camera in a device can be used to show how a person would look when using different cosmetics)

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7

Application	Most suitable input device
Entering text and numbers into a word processor or spreadsheet	Keyboard
Selecting an option or icon from an on-screen menu	Touch screen/pointing device
Input a user's voice into a computer as part of a voice recognition system	Microphone
Converting a hard copy document into an electronic form to be stored in a computer	(2D) scanner
Reading of a QR code using a smartphone or tablet	Camera
Keying in digits from a barcode which didn't scan correctly	Keypad
Read data directly from the surroundings, such as taking a temperature	Sensor

8

Touch screen technology	Advantages	Disadvantages
Capacitive	• better image clarity than other technologies • very durable screens with good scratch resistance • allows multi-touch facility	• surface capacitive screens only allow use of bare fingers or stylus • sensitive to electromagnetic radiation
Infrared	• allows multi-touch facility • good screen durability • not affected by scratched/cracked screen	• can be sensitive to water/moisture on screen surface • possible accidental activation • sensitive to light interference
Resistive	• good resistance to dust and water • can be used with fingers, stylus or gloved hand	• low touch sensitivity • doesn't support multi-touch facility • poor visibility in strong sunlight • vulnerable to scratches on screen

9 **a** Three from:

- higher contrast ratios
- greater longevity
- quieter running
- no issues lining up images
- smaller and lighter than LCD projectors
- better suited to dusty and smoky atmospheres

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b Two from:

- The image tends to suffer from 'shadowing' during moving images.
- They do not have grey components in the image.
- Colour definition is not as good as LCD projectors.

10 a inkjet printers:

advantages:

- don't produce ozone gas
- produce high-quality photo/vivid colour images
- small footprint
- don't need to warm up
- very low purchase price

disadvantages:

- have a small ink reservoir
- small paper trays
- not suitable for long print runs (ink cartridges can be used up quickly)
- can't use highlighter pens on printouts as ink smudges
- print heads are not very durable

laser printers:

advantages:

- have large paper trays for long print runs
- have large capacity toner cartridges
- very quiet printing
- cost per page is very low

disadvantages:

- produce ozone gas (an irritant) (Don't allow this point if given as an advantage of using an inkjet printer.)
- expensive to maintain (cartridges and other parts, such as the diffuser, are expensive)

b i laser printer

- large number of leaflets; therefore, need large amounts of paper and ink to do the print run

ii inkjet printer

- one off, therefore, speed not essential
- inkjet printers good at producing high definition photo-quality printouts

iii 3D printer

- items for sale need to be solid coins and feel like the real thing
- 3D printers can produce coins from layers of resin/plastic or metal powders resembling original coins

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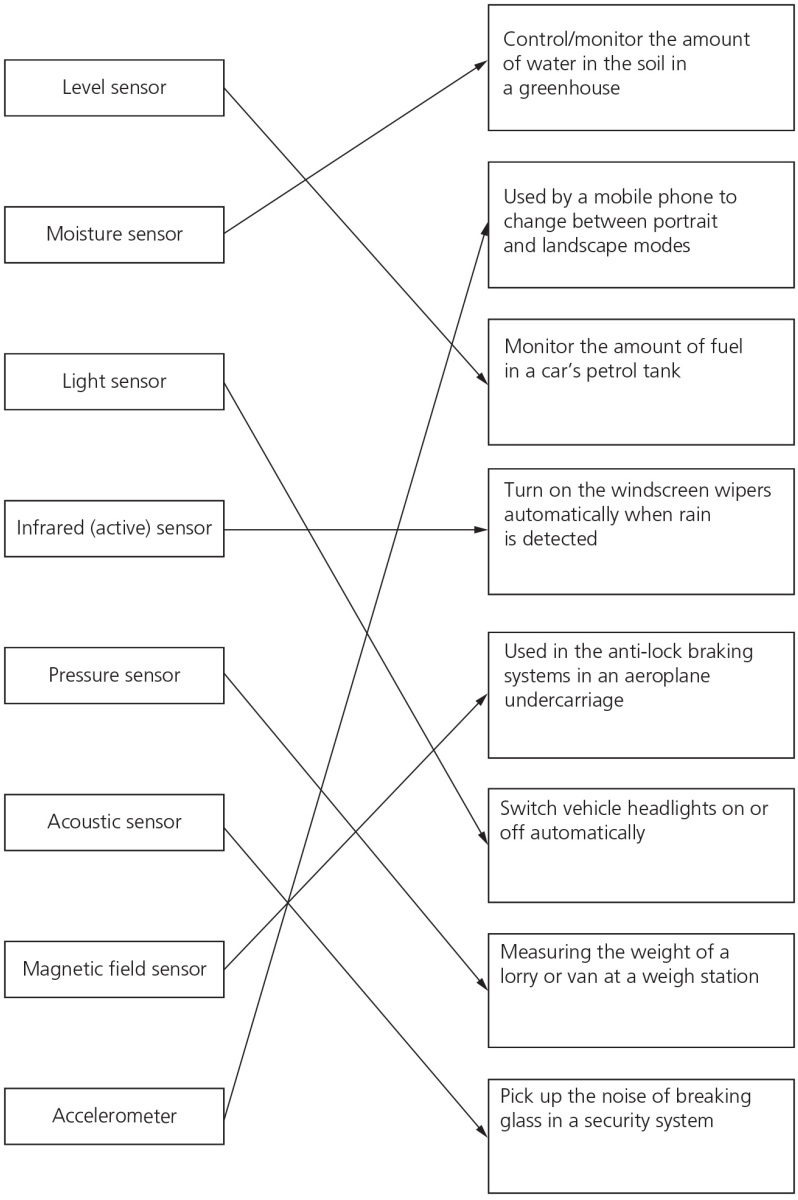
11 a

- The enthusiast makes a drawing of the part or finds a blueprint.
- The finalised drawing is imported into special software that prepares it in a format understood by the 3D printer.
- A solid object is built up layer by layer; the material used is a metallic powder in this case.
- The object is removed from the printer and is then prepared by removing any support material.

b Three from:

- prosthetic limbs
- facial construction and other reconstructive surgery
- making parts in the aerospace industry
- fashion and art
- prototypes, for example, in the design of a new games console

12



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13 a infrared sensor (passive and/or active), acoustic sensor (microphone), pressure sensor

b
- The sensors continually send data to the microprocessor.
- Data is converted to digital form by an ADC.
- The microprocessor compares data to pre-set values.
- If an infrared (active) beam is broken, the microprocessor sends a signal to an actuator to operate a siren/flashing lights.
- If an infrared (passive) value > pre-set value, the microprocessor sends a signal to an actuator to operate a siren/flashing lights.
- If the microphone/acoustic sensor readings > pre-set value, the microprocessor sends a signal to an actuator to operate a siren/flashing lights.
- If a pressure sensor reading > pre-set value, the microprocessor sends a signal to an actuator to operate a siren/flashing lights.
- The process is continuous.

14 a primary:
- directly addressable by the CPU
- part of the CPU/motherboard
- contains RAM/ROM/cache

secondary:
- not directly addressable by CPU
- non-volatile devices
- often external to computer
- for example, HDD, SSD

b

Description	**RAM (✓)**	**ROM (✓)**
Temporary memory device	✓	
Non-volatile memory		✓
Data stored in this memory cannot be altered		✓
Permanent memory device		✓
Stores data and part of the operating system currently in use	✓	
Can be increased in size to improve the operational speed of a computer	✓	

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c i
- radio frequency to control the device
- software to carry out the basic functions of a device
- start-up procedures when the device is powered up

ii
- a user's own programs/routines to control the car
- part of the software needed to control the car's manoeuvres
- temporary new radio frequencies

iii
- to allow connection to the computer to get software updates
- to connect a small memory stick containing pre-programmed routines

15

	Technology used		
Description	**Magnetic**	**Optical**	**Solid state**
Makes use of floating gate and control gate technology			✓
Disk surfaces are made up of tracks and sectors; storage relies on certain properties of the iron oxide coating	✓		
Devices using this technology have no moving parts			✓
Data is stored in 'pits' and 'lands' on a single spiral track running from the centre outwards		✓	
Disks use the properties of laser radiation to allow read and write operations		✓	
Data is stored by controlling the movement of electrons within NAND chips			✓
This technology is affected by strong magnetic and radio fields	✓		

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16

Statement	True (✓)	False (✓)
Devices using magnetic media have a very slow data transfer rate		✓
HDD platters can be made from ceramic, glass or aluminium coated in iron oxide	✓	
HDD surfaces are split up into tracks and sectors	✓	
HDD use a more reliable technology than solid state devices, such as SSD		✓
DVD-Rs are used to read and write data several times		✓
DVD and Blu-ray disks can both use dual-layer technology	✓	
SSDs suffer from a high degree of latency		✓
Solid-state devices, known as flash drives, use NAND chip technology	✓	
SSD data access time is approximately 100 times less than the data access time for HDD	✓	
Memory cards (such as the SD and XD cards used in cameras) are examples of magnetic media		✓

17 How VM works:

- It utilises RAM and HDD/SSD.
- RAM is physical memory and swap space on a HDD/SSD is called virtual memory (VM).
- VM management moves the oldest data out of RAM into swap space on a HDD/SSD.
- The oldest data is moved out to allow a new program to gain RAM access.
- The bit map is updated to reflect the new memory status.

Benefits:

- It gives the illusion of unlimited RAM storage/memory.
- Programs can be larger than physical memory and can still be executed.
- It provides more efficient multi-programming.

Drawbacks:

- There is memory wastage.
- There is the need to buy additional RAM.
- If a HDD is used, one drawback is disk thrashing.
- Disk thrashing can lead to HDD premature failure.

18 a It is a method of data storage where data is stored on offsite servers:

- There are three types: public, private and hybrid.
- The same data is often stored on more than one server in case of maintenance or repairs/backing up – this is called data redundancy.

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b public: storage environments where client and cloud storage provider are separate companies/organisations

private: storage provided by a dedicated environment behind a company firewall; client and cloud storage provider are integrated and act as a single entity

hybrid: a combination of public and private; some sensitive data stored on a private cloud whereas other data can be stored on a public cloud

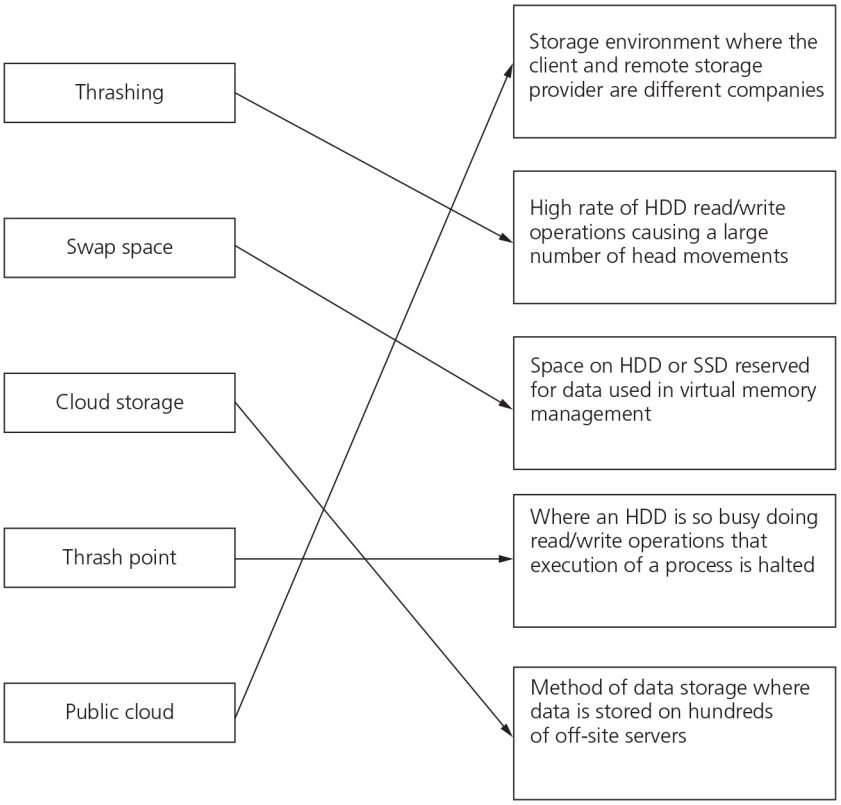
c **i** Three from:

- Customer/client files stored on the cloud can be accessed at any time from any device anywhere in the world provided internet access is available.
- There is no need for a customer/client to carry an external storage device with them, or even use the same computer to store and retrieve information.
- The cloud provides the user with remote back-up of data with obvious benefits to alleviate data loss/disaster recovery.
- If a customer/client has a failure of their hard disk or back-up device, cloud storage will allow recovery of their data.
- The cloud system offers almost unlimited storage capacity.

ii Two from:

- If the customer/client has a slow or unstable internet connection, they might have many problems accessing or downloading their data/files.
- Costs can be high if large storage capacity is required; it can also be expensive to pay for high download/upload data transfer limits with the customer/client internet service provider (ISP).
- The potential failure of the cloud storage company is always possible – this poses a risk of loss of all back-up data.

19



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20 a i NIC

ii MAC

iii IP

iv router

v firewall

b i It is a temporary IP address assigned to a device which changes each time a device logs onto the internet. It uses DHCP to assign the temporary IP addresses.

ii

Dynamic IP address	**Static IP address**
Greater privacy because it changes each time a user logs on	Because a static IP address doesn't change, it allows each device to be fully traceable
A dynamic IP adress can be an issue when using, for example, VoIP because this type of addressing is less reliable because it can disconnect and change the IP address causing the VoIP connection to fail	It allows for faster upload and download speeds
	It is more expensive to maintain because the device must be constantly running so that information is always available

c i 1 = computer

2 = switch

3 = router

4 = external network (e.g. internet)

ii
- It enables data packets to be routed between different networks, for example, joining a LAN to a WAN.
- A router takes data transmitted in one format from a network (using one protocol) and converts the data to a protocol and format understood by another network.
- A router inspects data packets sent to it from any computer or connected network.
- Because all computers on the same network have the same part of an IP address, the router can send data packets to the correct switch for onward delivery to the correct MAC address.

21 a internet service provider

b a set of rules agreed by devices using the internet

c
- IPv4 uses 32-bit addresses …
- … whereas IPv6 uses 128-bit addresses.
- for example: 190.188.0.250 (IPv4)
  fff0:a8fb:7a88:0fff:3d22:2088:66fb:f0fa (IPv6)
- There is less risk of IP address collision with IPv6.
- IPv6 has built-in authentication.
- IPv6 is a more modern/recent IP address protocol.

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